Tasks

Question 1

Design a class for Singly linked List named Node with data members **data** and **next** pointer. Your class must provide implementation for the following

- A default constructor for Node
- A parameterized constructor
- Getters for data and next
- Setters for data and next

Design a class for Singly linked List named SLinkedList with a pointer of type Node. This will carry the address of head initially, later it will be required for traversal.

Your class must provide implementation for the following

- A default constructor
- **Insert** function to insert data item. For every new data element to be inserted, you will have to reach up to the end point of LL and then perform insertion
  `void insert(T data)`
- **Insert to head** function to insert data element at the start of your LL
  `void insertAtHead(T data)`
- **isEmpty** function to check if there is no data element in the LL
  `bool isEmpty()`
- A **Search** function to search for any element provided by user
  `int search(T data)`
- **Update** function to update/replace one data element with another
  `void update(T value, T newValue)`
- A function **Insert at index** that will count data elements in the LL and then place a new data element in the specified location. To know exactly after which node we have to make insertion, we need count the elements and that will help us to identify index (index doesn't exist actually, but we are using this term to get to know the location for insertion)
  `void InsertAtIndex(T data, int index)`
- A function **Delete** to delete the specified data element, for this you will have to save the address of the next node that appears after the one to be deleted. Saving the address will help you after deleting the element to concatenate the list with the elements appearing after the element that is supposed to be deleted.
  `void remove(T value)`
- A function to **print** the data of LL
  `void print()`